KEMENTERIAN PENDIDIKAN, KEBUDAYAAN, RISET, DAN TEKNOLOGI
POLITEKNIK NEGERI MALANG
JURUSAN TEKNOLOGI INFORMASI
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Mata Kuliah	:	**PEMROGRAMAN BERBASIS FRAMEWORK (PBF)**
Program Studi	:	D4 – Teknik Informatika
Semester	:	6

Kelas	:	TI-3B
NIM	:	2341720164
Nama	:	Raudhil Firdaus Naufal
Jobsheet Ke-	:	1

Laporan Jobsheet

Langkah 1 - Pengecekan Lingkungan

Langkah	Jawaban/Deskripsi

1	node -v
	`C:\Users\raudh>node -v`
	`v24.11.0`
	Mengecek Versi Node JS dan mengecek jika sudah terinstall

2	Npm -v `C:\Users\raudh>npm -v` `11.6.1`, Mengecek Versi NPM yang terinstall

3	Git -V
	`C:\Users\raudh>git -v`
	`git version 2.43.0.windows.1`
	Mengecek apakah github sudah ada dan mengecek versinya

Langkah 2 - Membuat Project Next.js

Langkah	Jawaban/Deskripsi

1	Buat direktori baru dan masuk ke direktori kerja
	`D:\Matkul\Semester 6\FrameWork>mkdir praktikum`
	`D:\Matkul\Semester 6\FrameWork>cd praktikum`
	`D:\Matkul\Semester 6\FrameWork\praktikum>`
	Membuat folder baru dan memasuki folder baru tersebut

2	Jalankan perintah:
	`D:\Matkul\Semester 6\FrameWork\praktikum>npx create-next-app@13.4.7`
	`Need to install the following packages:`
	`create-next-app@13.4.7`
	`Ok to proceed? (y) y`
	`√ What is your project named? ... my-app`
	`√ Would you like to use TypeScript with this project? ... No / Yes`
	`√ Would you like to use ESLint with this project? ... No / Yes`
	`√ Would you like to use Tailwind CSS with this project? ... No / Yes`
	`√ Would you like to use `src/` directory with this project? ... No / Yes`
	`√ Use App Router (recommended)? ... No / Yes`
	`√ Would you like to customize the default import alias? ... No / Yes`
	`Creating a new Next.js app in D:\Matkul\Semester 6\FrameWork\praktikum\my-app.`
	`Using npm.`
	`Initializing project with template: default`
	`Installing dependencies:`
	`- react`
	`- react-dom`
	`- next`
	`- typescript`
	`- @types/react`
	`- @types/node`
	`- @types/react-dom`
	`- eslint`
	`- eslint-config-next`
	Membuat project dari node.js

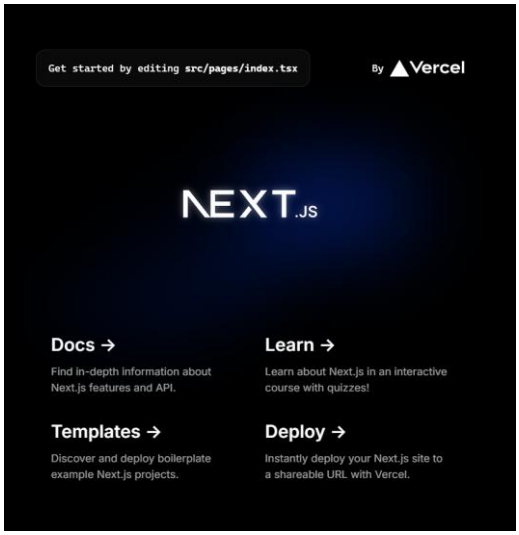
3	Masuk ke folder projectnya
	`D:\Matkul\Semester 6\FrameWork\praktikum>cd my-app`
	`D:\Matkul\Semester 6\FrameWork\praktikum\my-app>`

Langkah 3 – Menjalankan Server Development

Langkah	Jawaban/Deskripsi

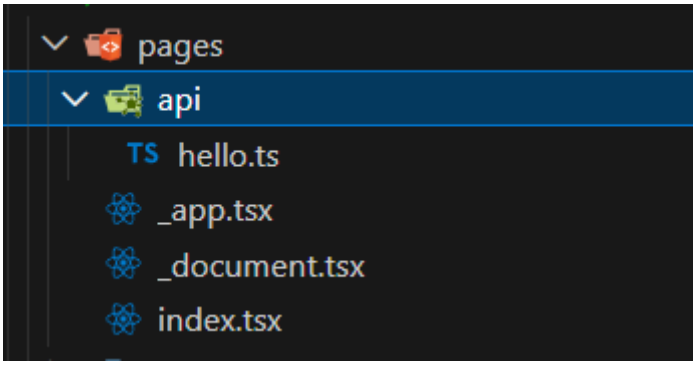
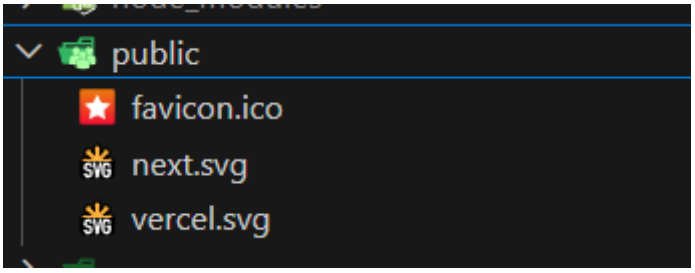
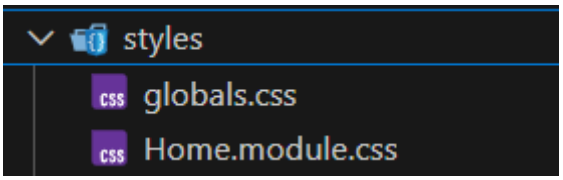
1	Masuk ke folder project:
	`D:\Matkul\Semester 6\FrameWork\praktikum>cd my-app`
	`D:\Matkul\Semester 6\FrameWork\praktikum\my-app>`

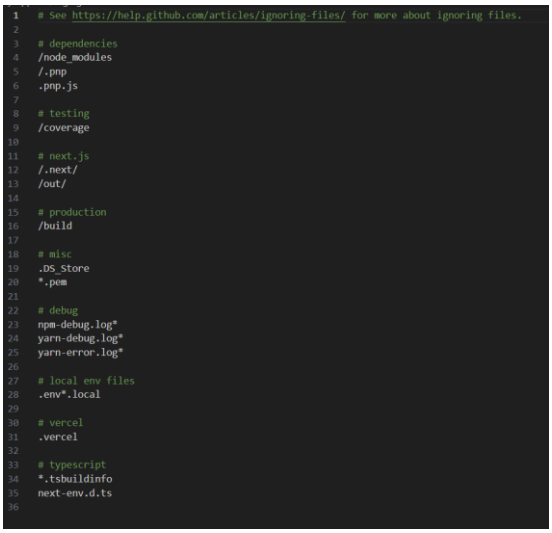
2	Jalankan aplikasi:
	`D:\Matkul\Semester 6\FrameWork\praktikum\my-app>npm run dev`
	`> my-app@0.1.0 dev`
	`> next dev`
	`▲ Next.js 16.1.6 (Turbopack)`
	`- Local: http://localhost:3000`
	`- Network: http://192.168.83.1:3000`
	`✓ Starting...`
	`Attention: Next.js now collects completely anonymous telemetry regarding usage.`
	`This information is used to shape Next.js' roadmap and prioritize features.`
	`You can learn more, including how to opt-out if you'd not like to participate in this anonymous program, by visiting the following URL:`
	`https://nextjs.org/telemetry`
	`We detected TypeScript in your project and reconfigured your tsconfig.json file for you.`
	`The following mandatory changes were made to your tsconfig.json:`
	`- jsx was set to react-jsx (next.js uses the React automatic runtime)`
	`✓ Ready in 1379ms`
	Menjalankan project yang telah dibuat sebelumnya

3	Buka browser dan akses: http://localhost:3000  Halaman default dari project yang telah dibuat

Langkah 4 – Mengenal Struktur Folder

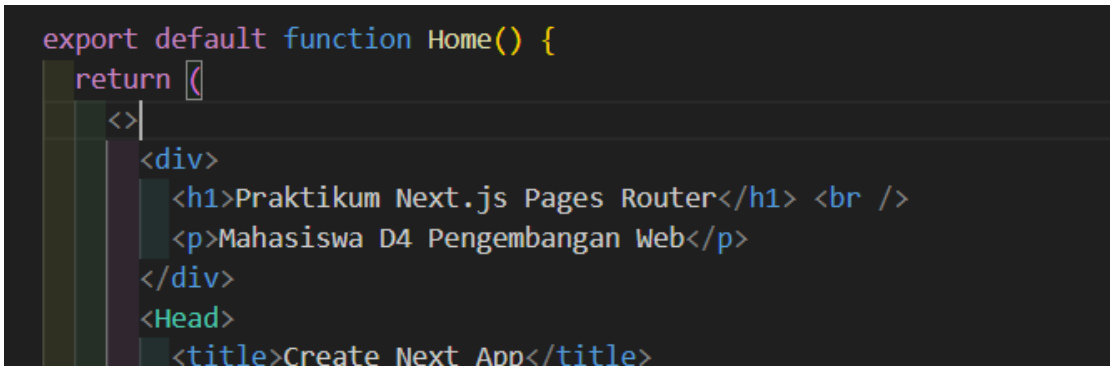
Langkah	Jawaban/Deskripsi

1	Amati folder utama:
	- pages/ → tempat routing halaman
	 
	
	- public/ → aset statis
	 
	
	- styles/ → file CSS
	 
	
	- package.json → konfigurasi project
	
	{
	"name": "my-app",
	"version": "0.1.0",
	"private": true,
	"scripts": {
	"dev": "next dev",
	"build": "next build",
	"start": "next start",
	"lint": "next lint"
	},
	"dependencies": {
	"@types/node": "25.3.0",
	"@types/react": "19.2.14",
	"@types/react-dom": "19.2.3",
	"eslint": "9.39.2",
	"eslint-config-next": "16.1.6",
	"next": "16.1.6",
	"react": "19.2.4",
	"react-dom": "19.2.4",
	"typescript": "5.9.3"
	}
	}

	- gitIgnore -> file konfigurasi di Git yang berfungsi untuk memberitahu Git file atau folder apa saja yang TIDAK perlu di-track / di-commit ke repository  ``` # See https://help.github.com/articles/ignoring-files/ for more about ignoring files. # dependencies /node_modules /.pnp .pnp.js # testing /coverage # next.js /.next/ /out/ # production /build # misc .DS_Store *.pem # debug npm-debug.log* yarn-debug.log* yarn-error.log* # local env files .env*.local # vercel .vercel # typescript *.tsbuildinfo next-env.d.ts ``` (1 # See https://help.github.com/articles/ignoring-files/ for more about ignoring files. 2 3 # dependencies 4 /node_modules 5 /.pnp 6 .pnp.js 7 8 # testing 9 /coverage 10 11 # next.js 12 /.next/ 13 /out/ 14 15 # production 16 /build 17 18 # misc 19 .DS_Store 20 *.pem 21 22 # debug 23 npm-debug.log* 24 yarn-debug.log* 25 yarn-error.log* 26 27 # local env files 28 .env*.local 29 30 # vercel 31 .vercel 32 33 # typescript 34 *.tsbuildinfo 35 next-env.d.ts 36)

Langkah 5 – Modifikasi Halaman Utama

Langkah	Jawaban/Deskripsi

1	Buka file pages/index.js, ubah isi halaman, misalnya:
	 
	``
	export default function Home() {
	return (
	<>
	<div>
	<h1>Praktikum Next.js Pages Router</h1>
	
	<p>Mahasiswa D4 Pengembangan Web</p>
	</div>
	<Head>
	<title>Create Next App</title>
	``
	Memodifikasi index.tsx dengan menambahkan Header dan paragraf

2	Simpan dan lihat perubahan di browser.

	 Perubahan yang terjadi

Langkah 6 – Modifikasi API

Langkah	Jawaban/Deskripsi

1	Buka folder api, Modifikasi hello.ts
	`// Next.js API route support: https://nextjs.org/docs/api-routes/introduction`
	`import type { NextApiRequest, NextApiResponse } from 'next'`
	
	`type Data = {`
	`  name: string`
	`  alamat: string`
	`}`
	
	`export default function handler(`
	`  req: NextApiRequest,`
	`  res: NextApiResponse<Data>`
	`) {`
	`  res.status(200).json({ name: 'John Doe', alamat: 'Jl. Merdeka No. 123' })`
	`}`
	
	Menambahkan data alamat dengan tipe data String dan menambahkan alamat pada json

2	Jalankan browser dengan Alamat http://localhost:3000/api/hello

	Pretty-print {"name":"John Doe","alamat":"Jl. Merdeka No. 123"} Json yang muncul setelah menambahkan alamat

3	Tambahkan extensions chrome, Jalankan browser chrome
	
	{
	"name": "John Doe",
	"alamat": "Jl. Merdeka No. 123"
	}
	Raw Parsed

Langkah 7 – Modifikasi Background

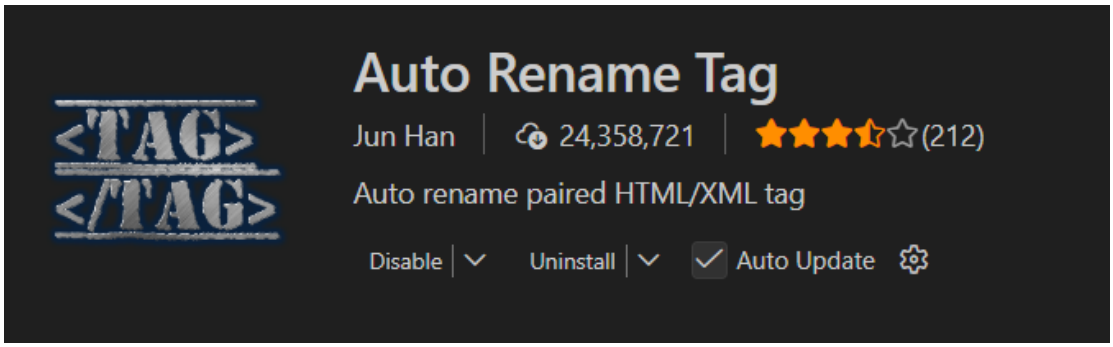
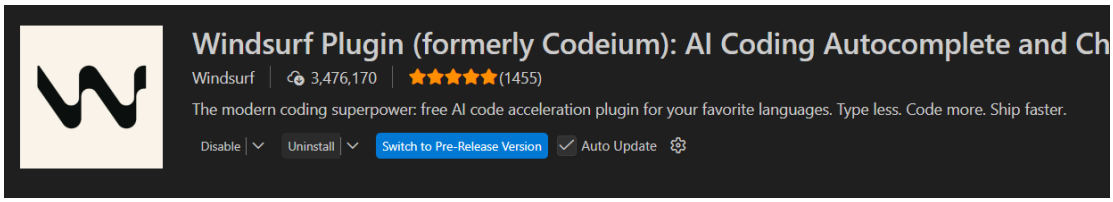
Langkah	Jawaban/Deskripsi

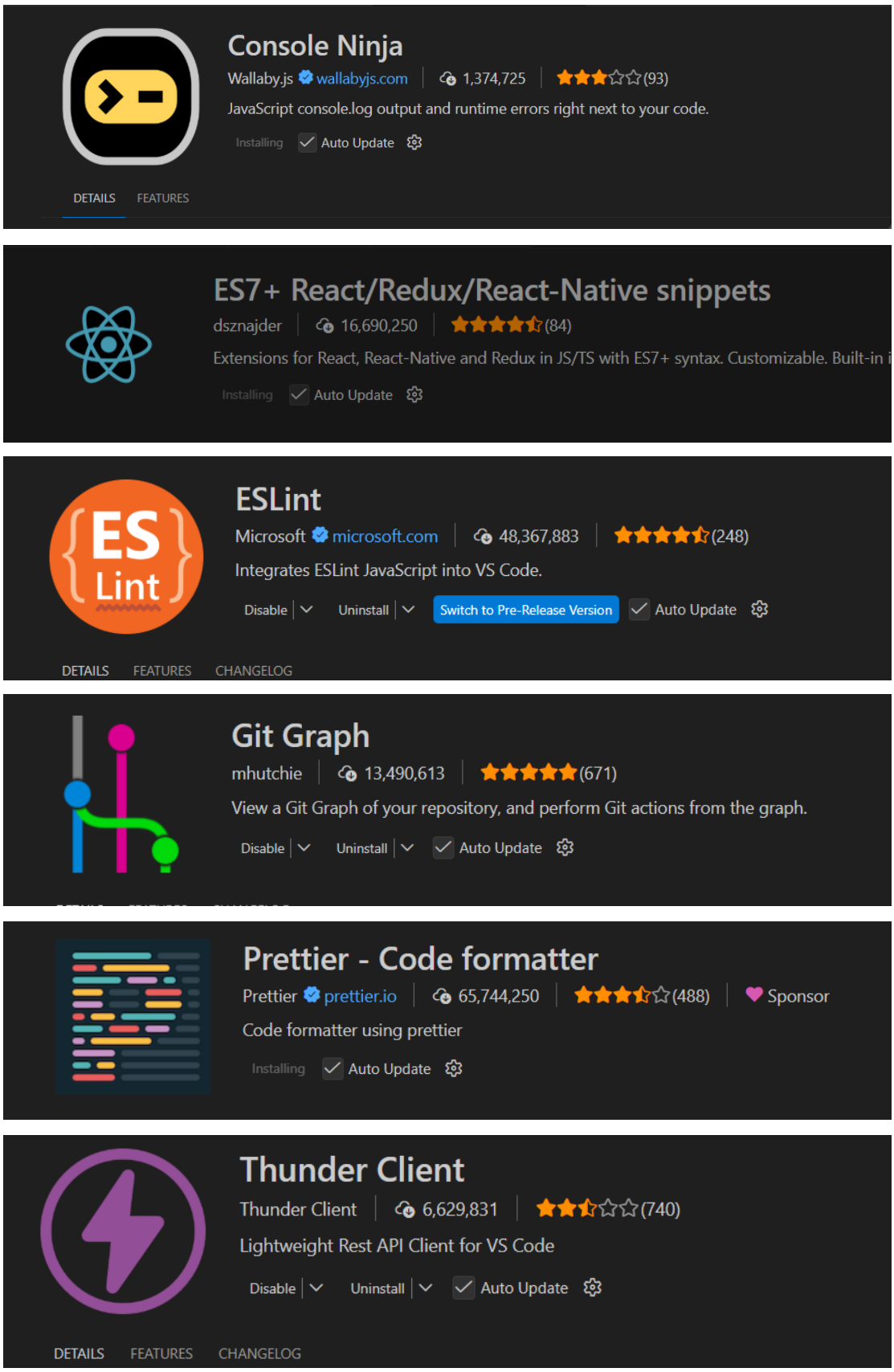
1	Buka file _app.tsx, Modifikasi menjadi berikut
	
	my-app > src > pages > _app.tsx > ...
	// import '@/styles/globals.css'
	import type { AppProps } from 'next/app' Ctrl+I Windsurf Command
	
	export default function App({ Component, pageProps }: AppProps) {
	return <Component {...pageProps} />
	}
	
	Melakukan command pada line import

2	Jalankan localhost

	 Halaman berubah saat import di command

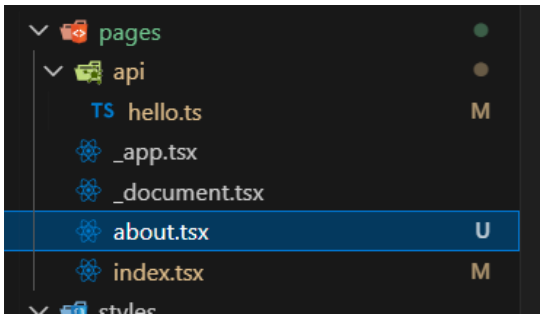
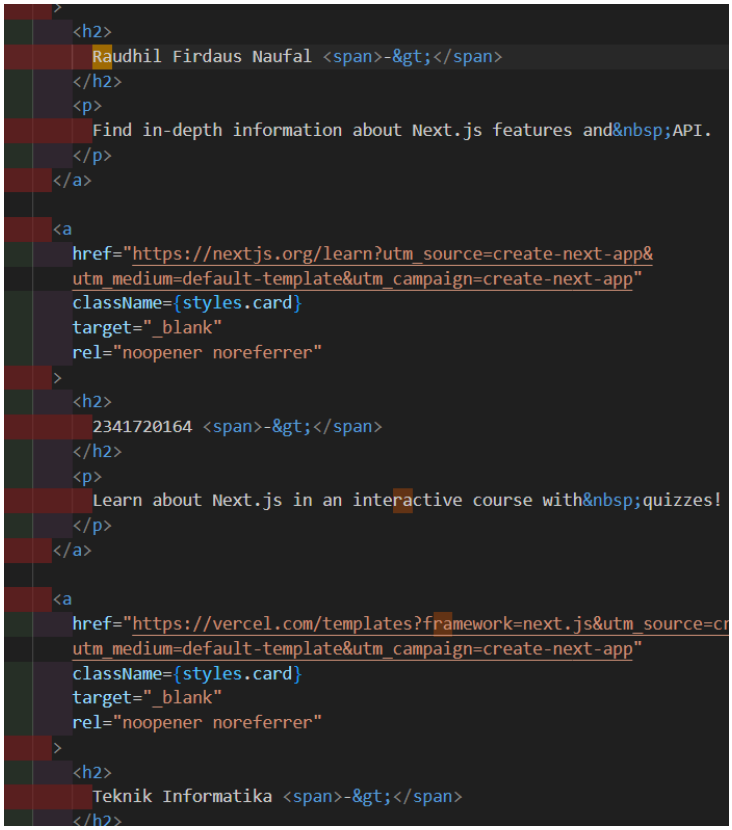
Langkah 8 – Setup ext pada VSCode

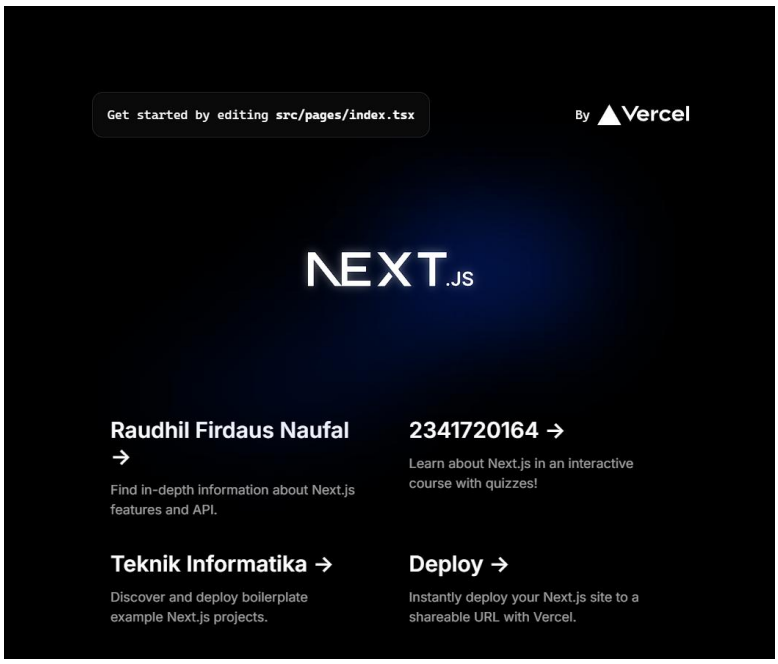
Langkah	Jawaban/Deskripsi
1	 


Console Ninja
Wallaby.js wallabyjs.com 1,374,725 (93)
JavaScript console.log output and runtime errors right next to your code.
Installing Auto Update
DETAILS FEATURES
ES7+ React/Redux/React-Native snippets
dsznajder 16,690,250 (84)
Extensions for React, React-Native and Redux in JS/TS with ES7+ syntax. Customizable. Built-in
Installing Auto Update
ESLint
Microsoft microsoft.com 48,367,883 (248)
Integrates ESLint JavaScript into VS Code.
Disable Uninstall Switch to Pre-Release Version Auto Update
DETAILS FEATURES CHANGELOG
Git Graph
mhutchie 13,490,613 (671)
View a Git Graph of your repository, and perform Git actions from the graph.
Disable Uninstall Auto Update
Prettier - Code formatter
Prettier prettier.io 65,744,250 (488) Sponsor
Code formatter using prettier
Installing Auto Update
Thunder Client
Thunder Client 6,629,831 (740)
Lightweight Rest API Client for VS Code
Disable Uninstall Auto Update
DETAILS FEATURES CHANGELOG

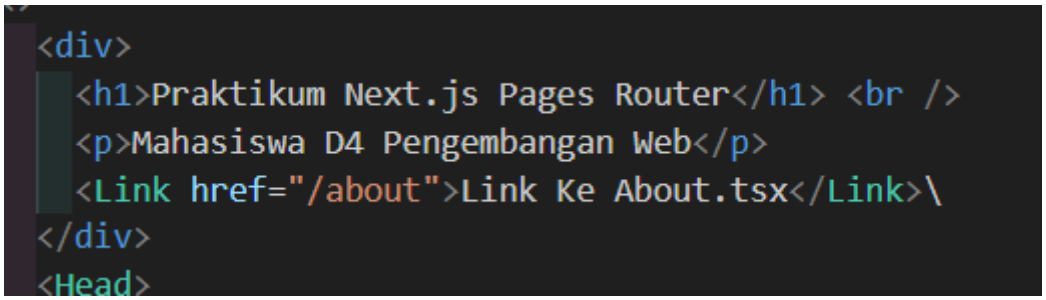
E. Tugas Praktikum

Tugas 1 (Wajib)

Langkah	Jawaban/Deskripsi
1	Buat halaman baru about.js di folder pages. 
2	Tampilkan: - Nama Mahasiswa - NIM - Program Studi 

Tugas 2 (Pengayaan)

Langkah	Jawaban/Deskripsi
1	Tambahkan minimal 1 link navigasi dari halaman utama ke halaman about. 

F. Pertanyaan Refleksi

Langkah	Jawaban/Deskripsi
1	Mengapa Pages Router disebut sebagai routing berbasis file? Karena file file yang berada di pages, dijadikan rute URL untuk diakses, sebagai contoh pada pages saya membuat about.tsx dan di route menjadi /about
2	Apa perbedaan Next.js dengan React standar (CRA)? Next.js: framework react yang menyediakan fitur siap pakai seperti Server-side Rendering, Static site generation dan file based routing untuk SEO yang unggul dan performa lebih cepat. CRA: Library UI Client-Side Rendering yang memerlukan konfigurasi tambahan untuk fitur serupa.
3	Apa fungsi perintah npm run dev? Npm run dev digunakan untuk menjalankan server pengembangan local pada proyek berbasis node.js
4	Apa perbedaan npm run dev dan run build ? Npm run dev: Digunakan untuk menjalankan server pengembangan local pada proyek berbasis node.js Npm run build: Digunakan untuk mengompilasi, meminifikasi, mengoptimalkan kode untuk lingkudan produksi. Jadi singkatnya dev untuk mengunakan kecepatan sedangkan build mengutamakan performa.

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